COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4 · MODULE 6

# Application Protocols: the Web, Email and DNS

1 · Network applications | 2 · The Web and HTTP | 3 · Email | 4 · DNS

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OUTLINE

# Module 6 — roadmap

- General concepts about network applications
- One network application: the Web
- HTTP: the protocol that supports the Web
- Email
- DNS: the Domain Name System

MODULE 6

# Network Applications

What an application needs from the layers beneath it

1 · Network applications | 2 · The Web and HTTP | 3 · Email | 4 · DNS

NETWORK APPLICATIONS

# What is a network application?

**Programs that run on different end systems** and communicate with each other over the network.

**Web browsing**

Google Chrome, Mozilla Firefox, Microsoft Edge

**Email**

Gmail, Outlook

**Instant messaging**

Slack, WhatsApp, Facebook Messenger

**Streaming services**

YouTube, Netflix

**Social media**

Instagram, TikTok

**AI assistants**

Claude, ChatGPT, Gemini

# Application-layer protocols

An **application-layer protocol** is a big piece of a network application.

It defines how the application's processes **pass messages**: which messages exist, their format, and when to respond — so e.g. any browser can talk to any web server.

| Network application | Application-layer protocol |
|---|---|
| Web application | HTTP |
| Email | SMTP |

To carry these messages, it relies on a **transport-layer protocol** (TCP or UDP).

# Choosing a transport protocol

- Application-layer protocols are controlled by the **developer**; transport-layer protocols are built into the **OS**.
- So the developer picks the OS transport protocol that best fits the application's needs:

| Property | Example applications | Transport protocol |
|---|---|---|
| **Reliable data transfer** | Email, IM, file transfer, financial apps | **TCP** preferred<br>UDP unsuitable |
| **Low delay** | Internet telephony, teleconferencing, multiplayer games | **UDP** preferred<br>TCP retransmissions add latency |

MODULE 6

# The Web and HTTP

The application-layer protocol behind every page you load

1 · Network applications | **2 · The Web and HTTP** | 3 · Email | 4 · DNS

THE WEB

# The World Wide Web

**THE WEB**

The **World Wide Web** (aka Web) is a **network application**: a global collection of documents and other resources shared over the Internet — like a large file storage system.

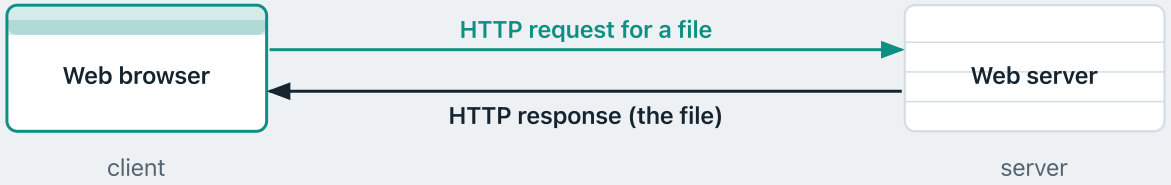
Each object on the Web (e.g., an image, an HTML file) is addressable by a **URL**:

```
http://www.someschool.edu/someDept/pic.gif
     └── host name ──┘ └──── path name ─────┘
```

# HTTP: the protocol behind the Web

**HTTP** | The **Hypertext Transfer Protocol** is the **application-layer protocol** that supports the Web — it defines how clients request objects by URL and how servers respond.

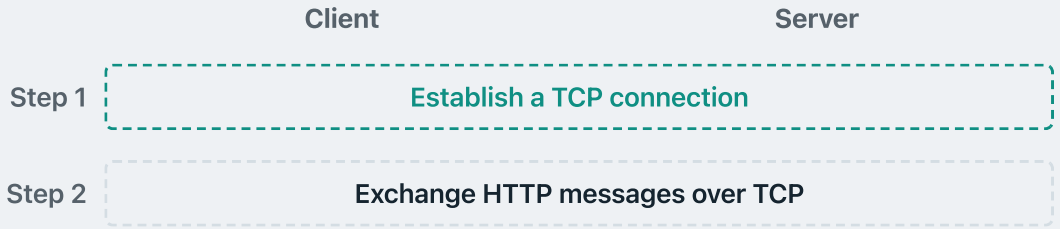
HTTP follows a **request–response** pattern: the client sends an **HTTP request** for an object, and the server answers with an **HTTP response** carrying it.



HTTP

# How HTTP works

HTTP uses **TCP** as its transport protocol — so before any HTTP message is sent, a TCP connection must be established first.




**STATELESS** HTTP is **stateless** — the server maintains no information about past client requests. Each request is handled independently.

# HTTP connections

### Non-persistent HTTP

- At most **one object** is sent over a TCP connection — connection is then closed
- HTTP/1.0 uses non-persistent HTTP

### Persistent HTTP

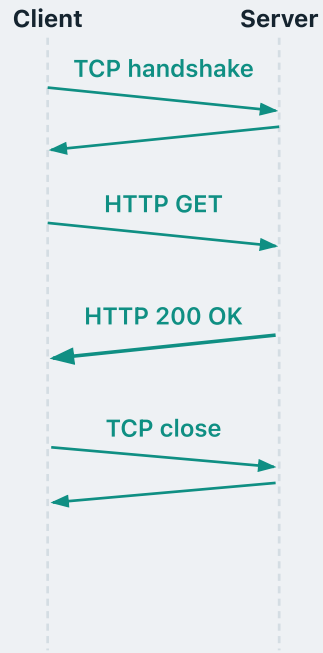
- **Multiple objects** can be sent over a single TCP connection
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP

| Protocol | Published | Connection type | Status |
|---|---|---|---|
| **HTTP/1.0** | 1996 | Non-persistent | Legacy |
| **HTTP/1.1** | 1997 | Persistent | Standard |
| **HTTP/2** | 2015 | Persistent | Standard |
| **HTTP/3** | 2022 | Persistent (QUIC) | Standard |

# Non-persistent HTTP in action

Suppose a client wants to retrieve a file from: `http://www.someSchool.edu/someDepartment/home.index`

1. Client establishes a **TCP connection** with the server
2. Client sends an **HTTP GET** request — path: `/someDepartment/home.index`
3. Server retrieves the HTML file, wraps it in an **HTTP response**, and sends it back
4. Server signals TCP to **close the connection**



# Response time: non-persistent HTTP

**RESPONSE TIME** — The time to retrieve a single object using HTTP — from the client's request until the object is received

**Q:** What is the response time for non-persistent HTTP?

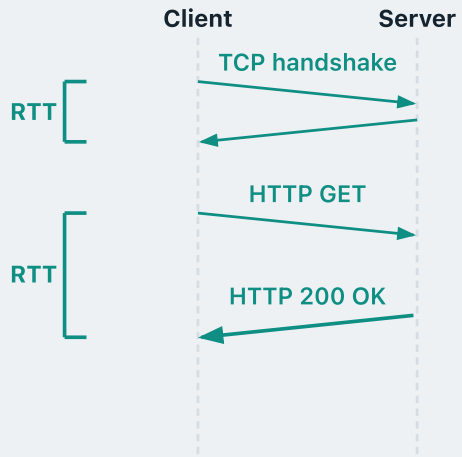
**A: 2 RTTs + file transmission time**

**RTT** — Recall: **round-trip time** — the time for a small packet to travel from client to server and back

**First RTT**: the TCP handshake — client sends SYN, server replies SYN-ACK

**Second RTT**: client sends HTTP GET, server sends back the response

With many referenced objects, this cost multiplies — a significant burden on the server



# Persistent HTTP in action

Client and server establish a TCP connection and exchange HTTP messages

Server **keeps the TCP connection open** after sending the first HTTP response

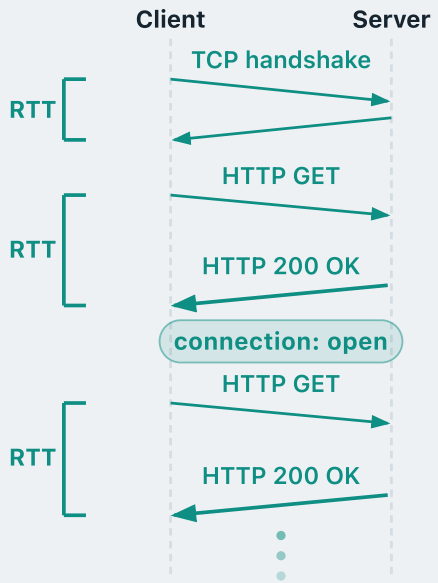
Subsequent HTTP messages between the same client and server use the **same TCP connection** — no new handshake needed

**Response time on one persistent connection:**

1 RTT for the initial TCP handshake, then 1 RTT for each HTTP request.

So for $n$ requests: **$\text{total} \approx (1 + n)$ RTT** (plus transmission time).

Amortised over the $n$ objects: $(1 + n)/n = (1/n + 1)$ RTT. As $n$ becomes large, $1/n \to 0$, so this approaches **~1 RTT per object**.

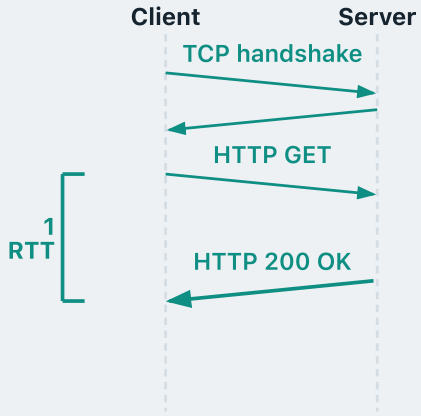



**Can we do even better than 1 RTT per object?**

# Improving HTTP efficiency further with pipelining

**WITHOUT PIPELINING**

Client issues a new request **only after** receiving the previous response — **1 RTT per file**



# Improving HTTP efficiency further with pipelining

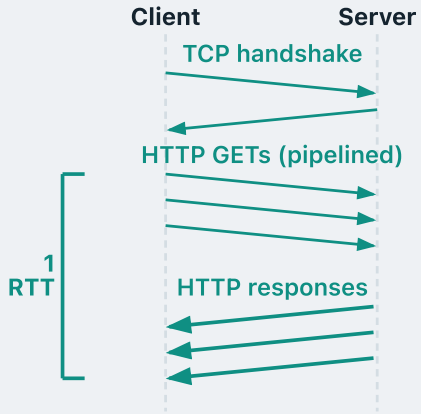
**WITHOUT PIPELINING**

Client issues a new request **only after** receiving the previous response — **1 RTT per file**

**WITH PIPELINING**

Client sends requests **as soon as it needs to retrieve an object**, without waiting for responses — e.g. in this example we pipeline 3 HTTP requests all together

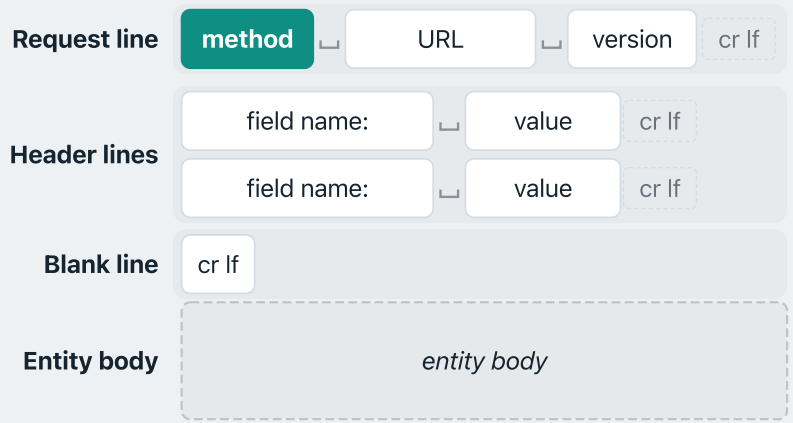
**Benefit**: pipelining 3 requests → response time per file ≈ **1 RTT / 3**



# HTTP request message format

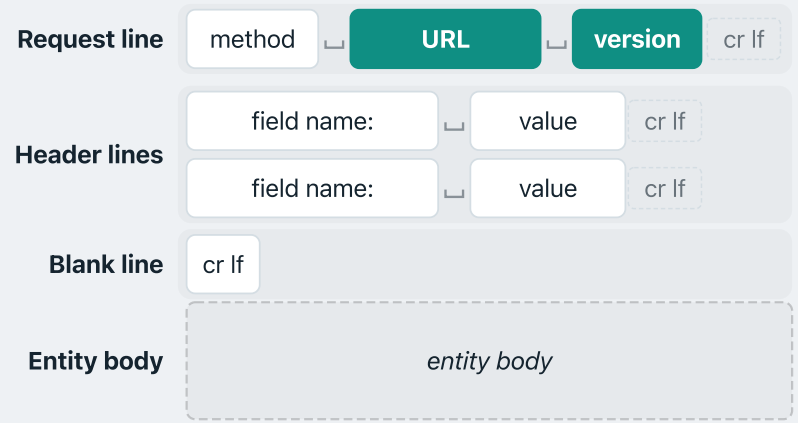
**Method — what action to perform**

| Method | Action |
|---|---|
| **GET** | **Retrieve a resource** |
| HEAD | Retrieve headers only (no body) |
| **POST** | **Submit data to a resource** |
| PUT | Store / replace a resource |
| **DELETE** | **Remove a resource** |



# HTTP request message format

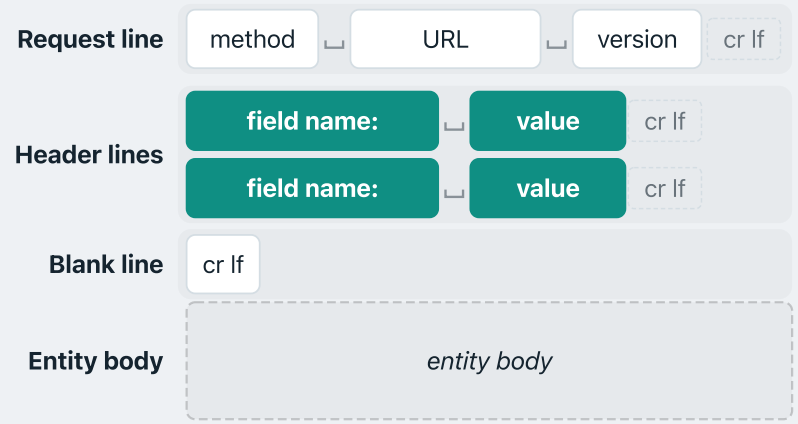
**URL**: path to the requested resource — e.g. `/someDept/pic.gif`
**HTTP Version**: identifies the protocol in use — `HTTP/1.0` or `HTTP/1.1`



# HTTP request message format

**Header lines**

- Zero or more `field-name: value` pairs, one per line
- They can appear in **any order**: HTTP is plain text, lines separated by CR LF — unlike IP and TCP headers, whose fields sit at fixed byte positions



# HTTP request message format

### Header lines

- Zero or more `field-name: value` pairs, one per line
- Common examples:

```
Host: www.example.com
```

Mandatory in HTTP/1.1 — identifies the target server

```
Content-Type: text/html; charset=UTF-8
```

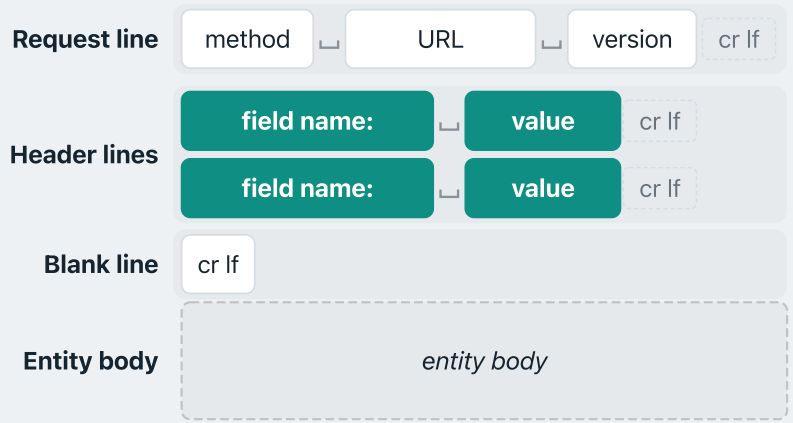
Format of the message body

```
Connection: close
```

Close the TCP connection after this response

```
Accept-Language: da, en-gb;q=0.8,
en;q=0.7
```

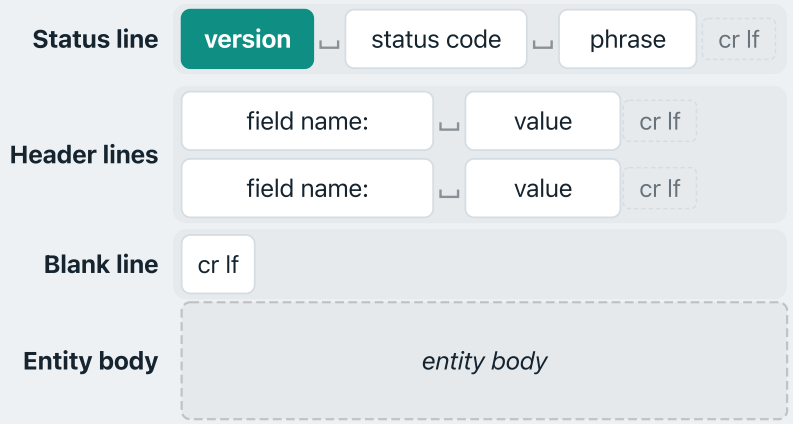
Preferred languages, weighted by q (default 1): Danish → British English → any English



# HTTP response message format

**Version**

The HTTP version the server is using — e.g. `HTTP/1.1`, just as in the request line.

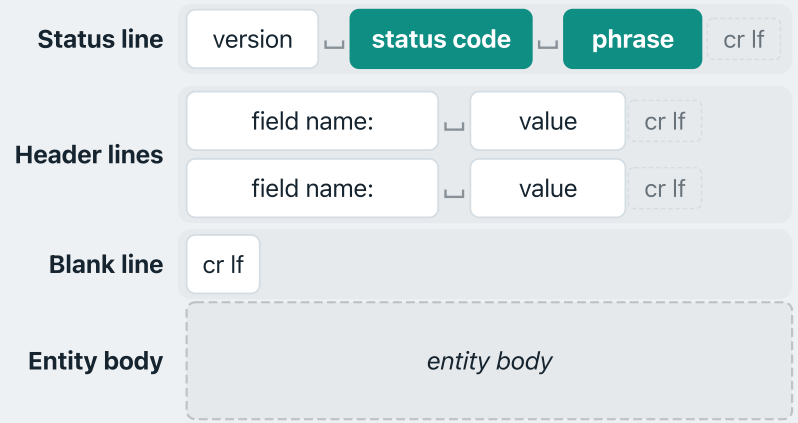


HTTP

# HTTP response message format

**Status code & phrase**

- **Status code**: 3-digit number indicating the result of the request
- **Phrase**: human-readable text accompanying the code — e.g. `OK`, `Not Found`

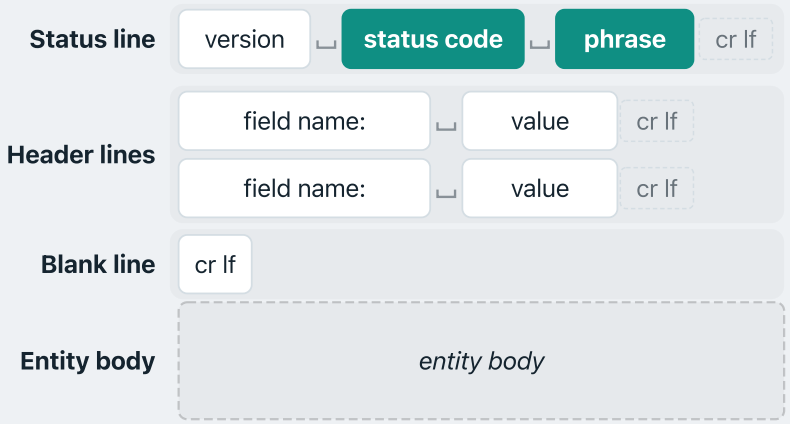


HTTP

# HTTP response message format

**Common examples**

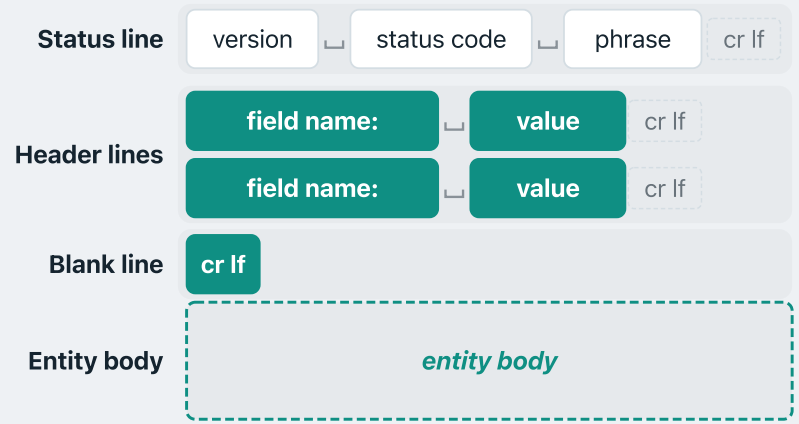
| Code | Phrase |
|---|---|
| 200 | OK |
| 301 | Moved Permanently |
| 400 | Bad Request |
| 404 | Not Found |
| 505 | HTTP Version Not Supported |



# HTTP response message format

**Header lines, blank line, entity body**

Same structure as in the request: `field-name: value` header lines, a blank line (a lone CRLF), then the body — here, the **requested object** itself.



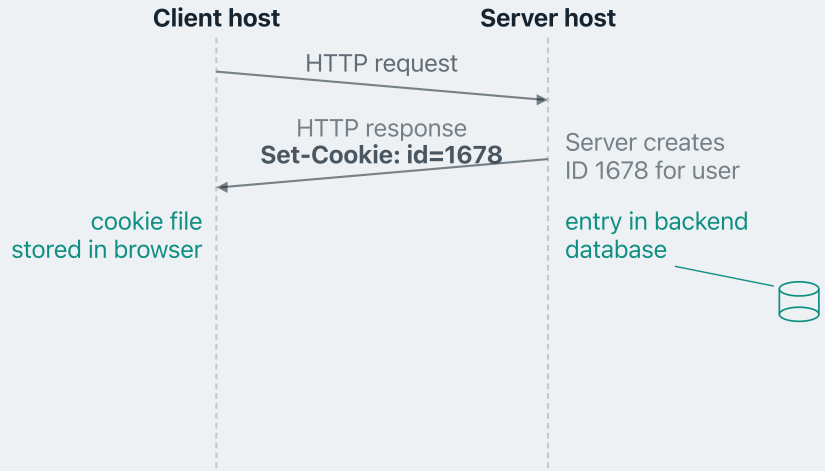
HTTP

# Cookies

> HTTP is **stateless** — every request is independent; the server retains no memory of past requests.
>
> So the server does not remember your earlier conversations with it.
>
> Yet many applications *need* that memory: e.g. you log in to Learn once; some time later, you visit Learn again — a **new HTTP request** — and it recognises you without asking you to log in.
>
> **How does the server know it is you?**

The answer: **cookies**.

# Cookies: an example [RFC 6265]




The client sends an HTTP request to the host, e.g. to log in

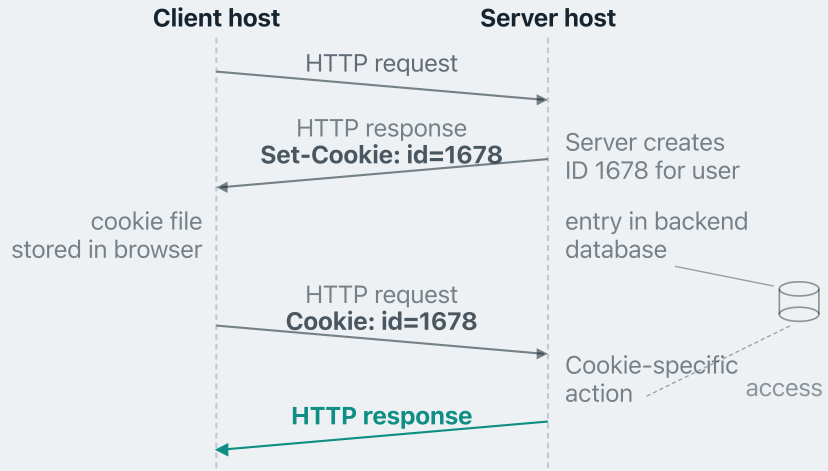
The server generates an **HTTP response**; on top of that, it prepares a **cookie** for the user.

The cookie uniquely identifies this session with the client, e.g. **id=1678**.

It is embedded in the response as a header line: **Set-Cookie: id=1678**.

The ID is stored as an entry in the server's **back-end database**, and in the client's **cookie file**, managed by the browser

# Cookies: an example [RFC 6265]




Later, the client sends another **HTTP request** to the same server.

The browser finds the cookie for this server in its cookie file,

and attaches it to the request as a header line: **Cookie: id=1678**.

The server uses the ID as a key to look up the user's state in its database

The server can then generate a **tailored response**, e.g. your logged-in Learn page, without asking you to log in again

HTTP

# One problem with the Web

Picture a university: thousands of users fetch the same popular pages and images, and every request travels across the Internet to the **origin server** that hosts the object.

Every one of those requests lands on the same origin server, through the same congested links — enough of them can **overwhelm** it.

**How can we avoid this?**

# Web caching

A **web cache** (also called a **proxy server**) is a network entity that satisfies HTTP requests on behalf of an origin server — storing copies of recently requested objects locally.




### Acts as a server

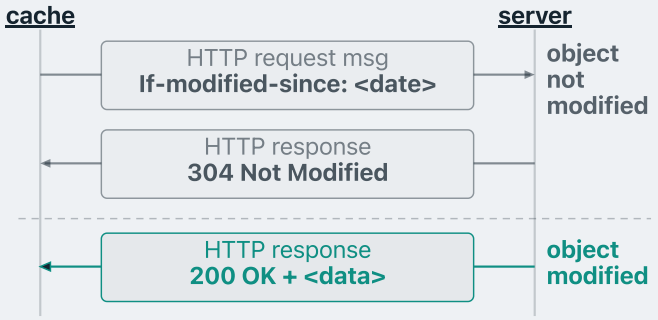
Receives HTTP requests from clients and, if it holds a copy of the object, returns it in the response — just like an origin server would

### Acts as a client

When the object is not cached, it issues its own HTTP request to the origin server, stores the copy, then answers the client

Requests answered by the cache get a **quicker response**, and caching **reduces how often** requests have to go to the origin server.

# Conditional GET




A **conditional GET** is an ordinary GET with an `If-modified-since` header line. A proxy uses it to check whether its cached copy is still up to date, e.g.:

```
GET /index.html HTTP/1.1
Host: www.example.com
If-modified-since: Mon, 10 Aug 2026 09:00:00 GMT
```

If the object is **unchanged**, server replies:

```
304 Not Modified
```

Object **changed**: server replies `200 OK` with the updated object in the body

MODULE 6

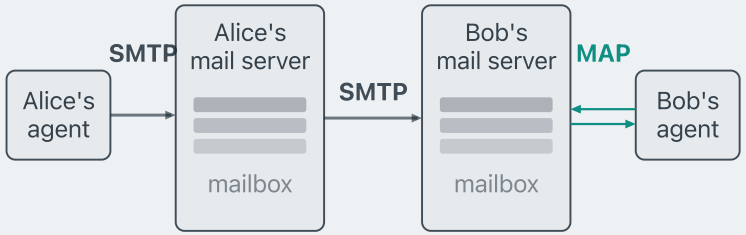
# Email

Electronic Mail

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EMAIL

# An overview of the email system




**User agent (UA)** — the app used to read, compose and send email (e.g. Outlook, Gmail, Apple Mail)

**Mail server** — a dedicated host that handles email on behalf of its users. It stores a **mailbox** per recipient for incoming mails and a **send queue** of outgoing mails

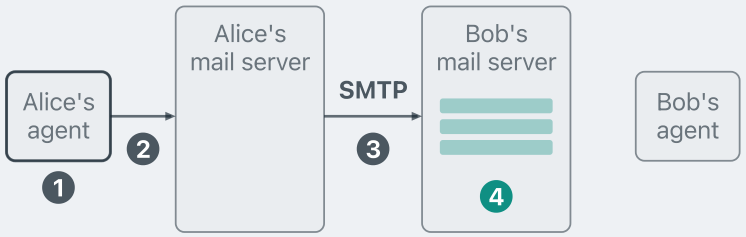
**SMTP** — Simple Mail Transfer Protocol; the sender's UA uses it to send a message to its own mail server

The sender's mail server then uses **SMTP** to transfer the message to the recipient's mail server across the internet

**Mail access protocol (MAP)** — used by UAs to retrieve messages from their mail server

# SMTP in action

SMTP has a **client side** and a **server side**, and always runs over **TCP**.




① Alice uses her UA to **compose** the message

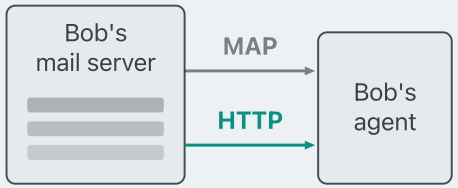
② Alice's UA (**SMTP client**) **sends** the message to her mail server (**SMTP server**)

③ The **SMTP client** on Alice's server opens a TCP connection to Bob's mail server and **transfers** Alice's message over it

④ The **SMTP server** on Bob's server receives the message and places it in Bob's **mailbox**

EMAIL

# Retrieving email from the server



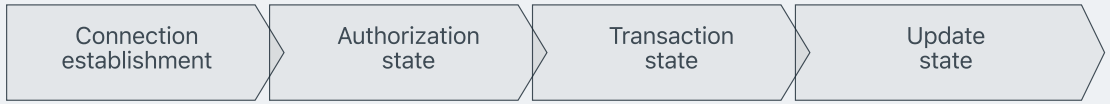

The recipient's UA retrieves email from its mail server either with a **mail access protocol (MAP)**, e.g.

- **POP3** — Post Office Protocol version 3 [RFC 1939]
- **IMAP** — Internet Message Access Protocol [RFC 9051]

or through **web-based email**, where the UA is a web browser that uses **HTTP** to reach the mail server

POP3

# How POP3 works




| Connection establishment | Authorization state | Transaction state | Update state |
|---|---|---|---|
| UA establishes a **TCP connection** to the mail server on **port 110** | UA sends **username** and **password** (cleartext) to authenticate | UA **retrieves messages**; can mark for deletion, remove marks, obtain statistics | Server **deletes** marked messages and **closes** the connection |

> Messages wait on the server until the UA downloads them

**Notable limitations:**

- Keeps no state on the server (e.g. read or unread), so what you do on one device is not seen on the others
- Does not support server-side folder management — users cannot organise emails into folders on the server

IMAP

# Internet Message Access Protocol

IMAP addresses POP3's limitations:

> IMAP **keeps per-message state on the server**, e.g. read or unread, flagged, so every device sees the same status

> It supports **folders on the server**: create folders and move messages into them, visible from any device

WEB-BASED EMAIL

# Accessing email via a browser

**Web-based email is accessed through a web browser**

The mailbox stays on the server; reading an email is like browsing the Web — the browser fetches each message as a web page

**Which application-layer protocol does the browser use to talk to the mail server?**

**HTTP**, the same protocol behind every other web page

**Web-based email services**

Examples: **Gmail**, **Outlook** (web)

EMAIL

# Comparing POP3, IMAP and web-based email

| | POP3 | IMAP | Web-based email |
|---|---|---|---|
| **Accessed with** | A mail app | A mail app | A web browser, over **HTTP** |
| **Messages** | Downloaded to the device; deleted from the server, or optionally kept | Kept on the server | Kept on the server |
| **Folders** | None on the server; a single mailbox | Created and kept on the server | Created and kept on the server |
| **Multiple devices** | Not synced: read status and folders exist only on each device | Read status and folders stay in sync across devices | Any device with a browser sees the same mailbox |

MESSAGE FORMAT

# Structure of an email message

An email message, as carried by **SMTP**, consists of **header fields**, followed (optionally) by a **body** [RFC 5322]

**Header fields** describe the message, e.g. `From:` and `Subject:`

The **body** carries the message content itself

SMTP requires the whole message to be **ASCII text**, which encodes English letters, digits and punctuation

**Problem:** ASCII alone cannot carry images, attachments, or non-ASCII characters (e.g. Chinese, Arabic)

**Header fields**

```
Date: Mon, 10 Aug 2026 09:00 GMT
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
Cc: charlie@example.com
```

**Body**

```
Hello Bob, here is the picture
you asked for. Enjoy!
```

MIME

# Multipurpose Internet Mail Extensions

**MIME** extends the message format with **additional header fields**

These fields declare the content's **type** and how it is **encoded**

so the receiver knows how to **decode the body**

e.g. **base64** encoding converts binary data into printable ASCII text, which can be used for image attachments

```
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
MIME-Version: 1.0
Content-Transfer-Encoding: base64
Content-Type: image/jpeg
---
(base64 encoded data…
…base64 encoded data)
```

→



decoded image

MODULE 6

# DNS

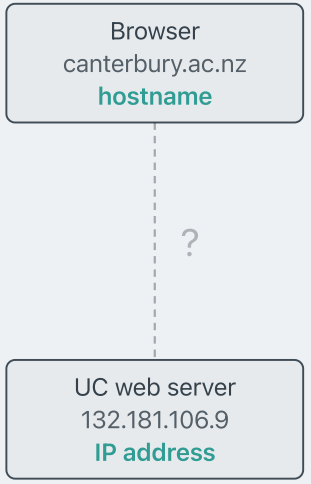
Domain Name System

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DNS

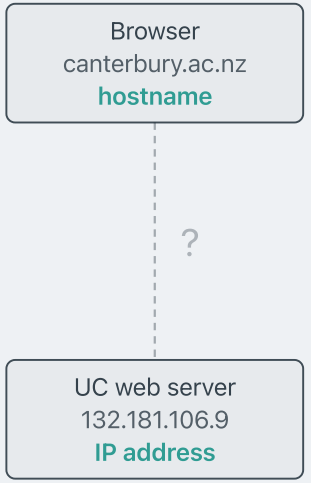
# Hostnames and IP addresses

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers, e.g. `132.181.106.9`
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?



DNS

# Domain Name System — overview



DNS

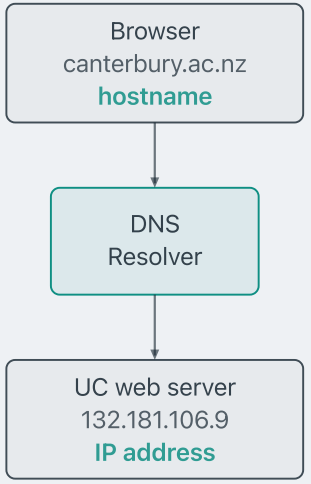
# Domain Name System — overview

**DNS** is that translation mechanism — it maps human-readable **hostnames** to numerical **IP addresses**

DNS stores this mapping in a **distributed database**, spread across a hierarchy of DNS servers [RFC 1034, 1035]

DNS is an **application-layer protocol** running mainly over **UDP on port 53** (TCP port 53 for large responses and zone transfers)

Why UDP? A query and its reply each fit in one small datagram, so there is no connection setup delay, providing a quick response.



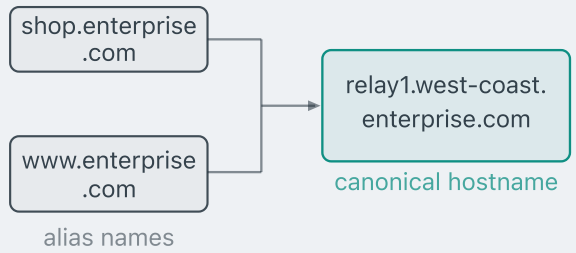

hostname → IP address

DNS SERVICES

# Additional services provided by DNS

**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember.

With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.



DNS SERVICES

# Additional services provided by DNS

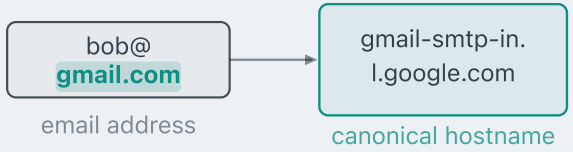
**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember.

With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.

**Mail server aliasing** — the same idea for email.

e.g. the mail server that handles Gmail has the cumbersome canonical hostname `gmail-smtp-in.l.google.com`.

Yet Bob's email address can simply be `bob@gmail.com`: DNS uses the part after the @, `gmail.com`, to find that mail server.



# Additional services by DNS: load distribution

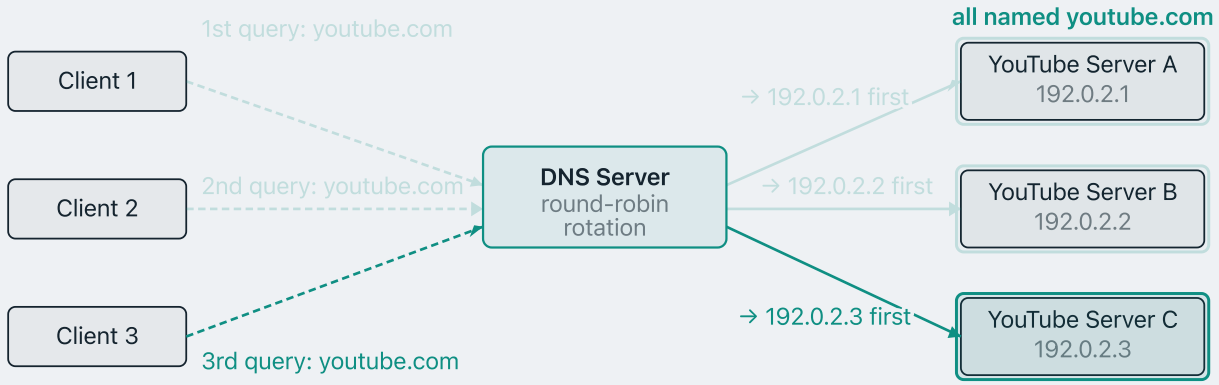
Many popular web services — e.g. YouTube, Google, Netflix — are hit by **millions of users** simultaneously. A **single server** cannot handle that volume.

The service is therefore distributed across **many server instances**, each with a **distinct IP address**, all sharing the **same hostname** (e.g. `youtube.com`).

### How are users assigned to these servers?

**DNS load balancing** — DNS stores all the IPs and **rotates their order** on each response, so consecutive clients get a different IP listed first.

e.g. three clients query `youtube.com` in turn:



DNS

# DNS is distributed, not centralised

Recall that DNS stores the hostname-to-IP mapping in a **distributed database**, spread across many DNS servers.

**Why not a centralised design — a single server hosting the whole database, so that every DNS query goes to it?**

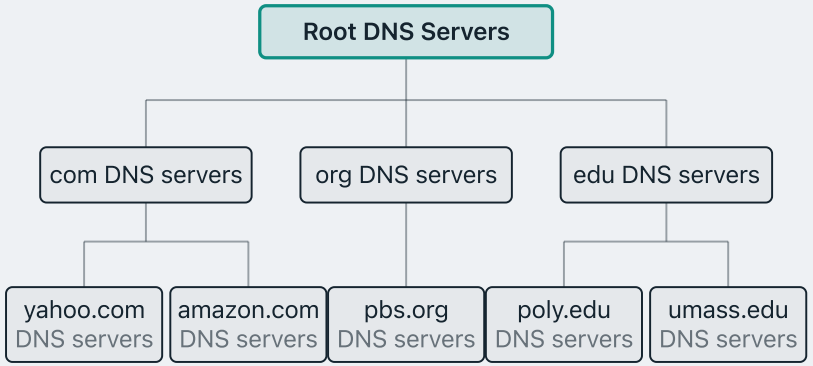
**Single point of failure**
If the one server crashes, DNS lookup fails for the entire Internet

**Traffic volume**
Billions of DNS queries per day — no single machine can handle that load

**Distant centralised database**
All queries must travel to one location — users far away face high latency on every lookup

**Maintenance**
Billions of hostname records change constantly — impossible to keep one database current

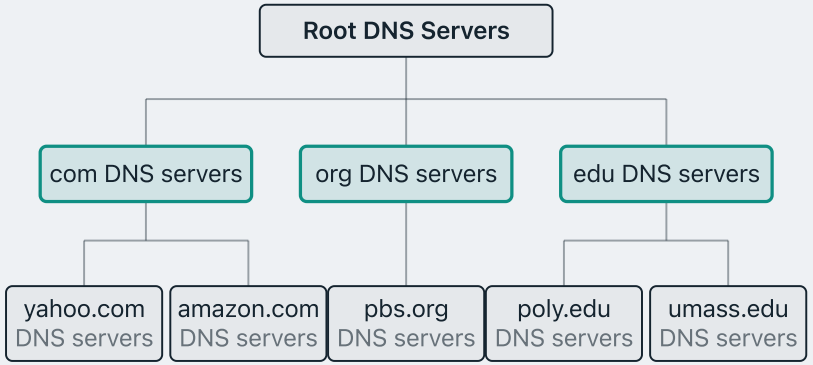
# A hierarchical distributed database




**Root DNS Servers**

- The top of the hierarchy: looking up a hostname from scratch **starts here**
- There are 13 root server identities (a–m.root-servers.net)
- Each is replicated: its IP address is shared by many machines worldwide, and a query goes to the nearest one

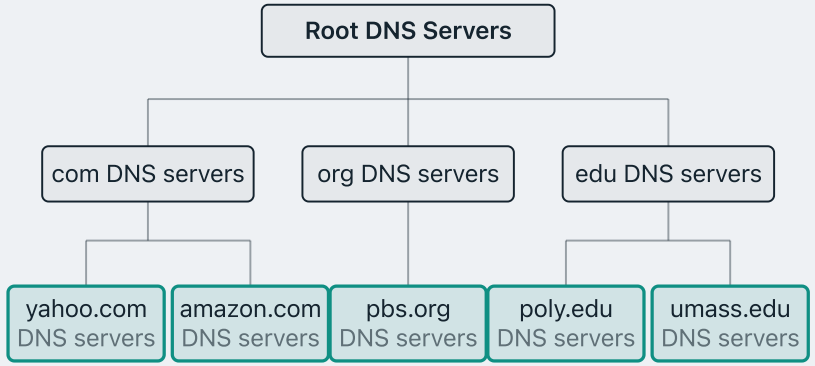
# A hierarchical distributed database




**Top-Level Domain (TLD) DNS Servers**

- Handle generic domains: **.com**, **.org**, **.edu**
- ...and country-code domains: **.nz**, **.uk**
- Each is run by a dedicated registry, e.g. **Verisign** (.com), **InternetNZ** (.nz)

# A hierarchical distributed database




**Authoritative DNS Servers**

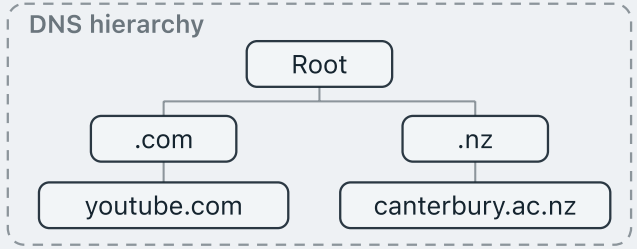
- Each organisation's own DNS server
- It stores the organisation's own records — the original, **definitive** copy that every answer for its hostnames comes from
- e.g. Canterbury's authoritative server holds the IP for `www.canterbury.ac.nz`

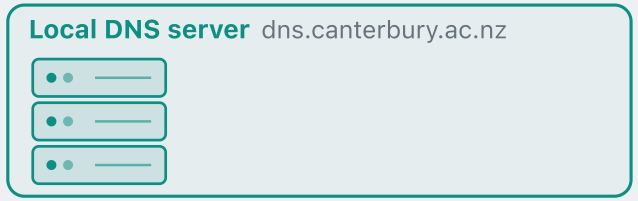
DNS ARCHITECTURE

# Local DNS server and DNS caching

Every ISP or institution operates a **local DNS server** (also called a *default name server*).

It is **not** part of the root / TLD / authoritative hierarchy — it sits **separately** from that.

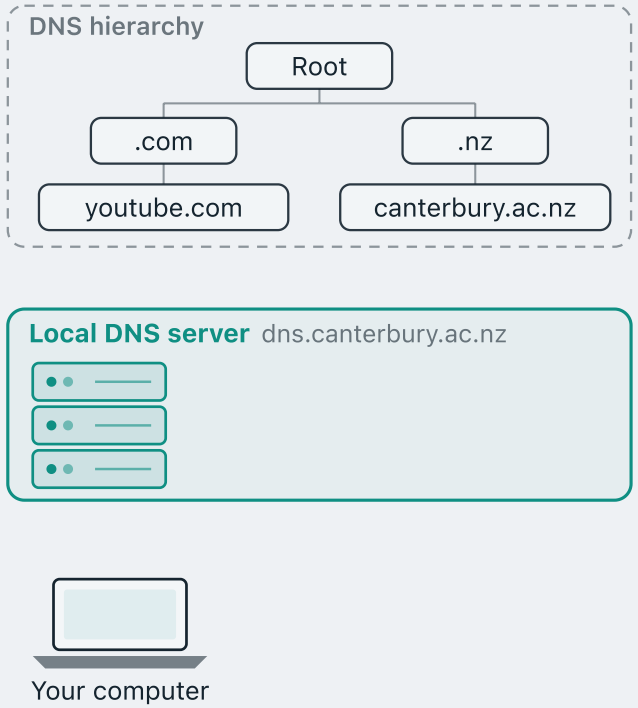


DNS ARCHITECTURE

# Local DNS server and DNS caching

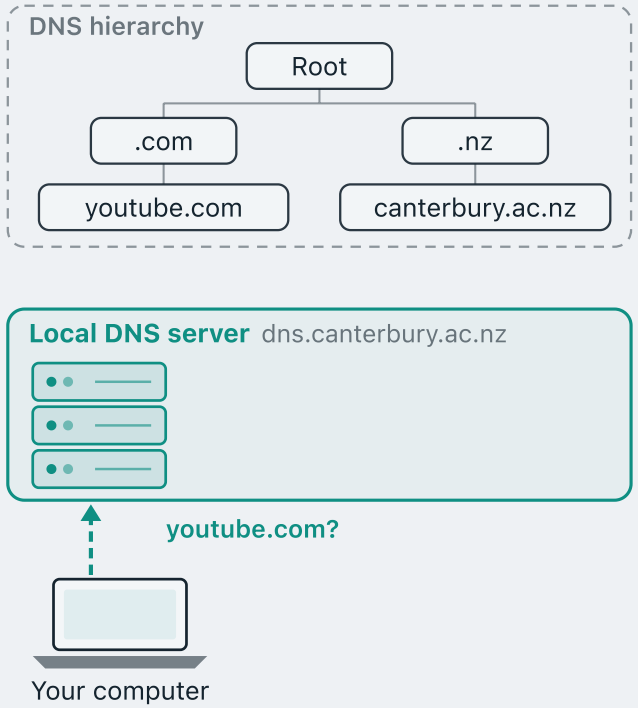
When you join the network, your device is automatically configured to use one — e.g. the University of Canterbury runs its own local DNS server for all campus users.



DNS ARCHITECTURE

# Local DNS server and DNS caching

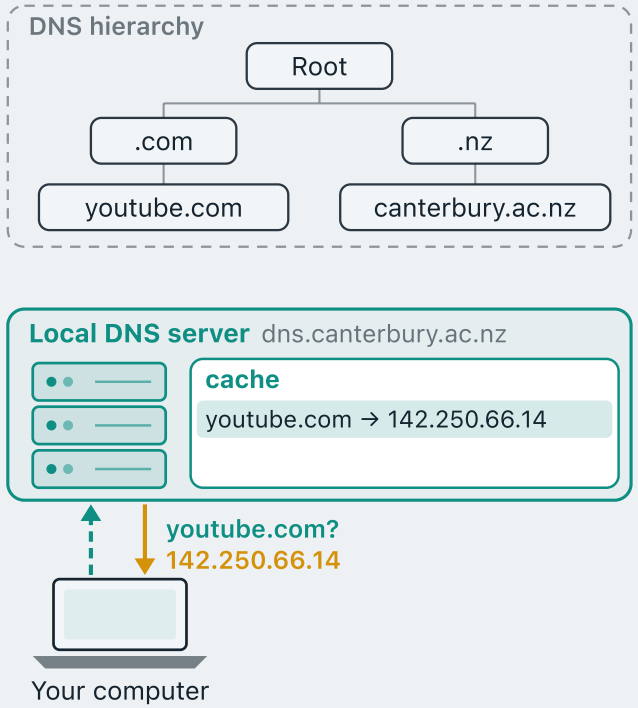
When your machine makes a DNS lookup, the query goes **first to the local DNS server**.



DNS ARCHITECTURE

# Local DNS server and DNS caching

**If the answer is cached:** the local DNS server replies **immediately** — no hierarchy lookup needed.



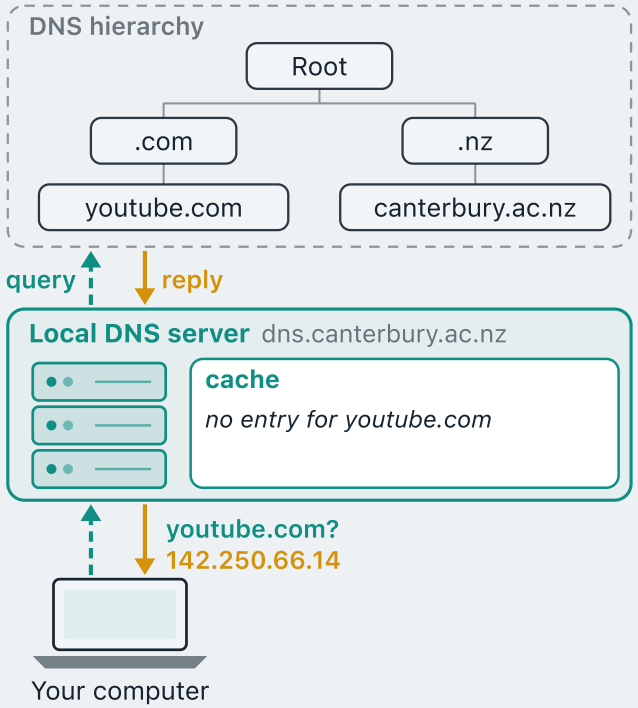
DNS ARCHITECTURE

# Local DNS server and DNS caching

**If not cached:** the local server **forwards the query into the hierarchy**.

The hierarchy replies with the IP address.

The local server **relays the reply** back to you,



DNS ARCHITECTURE

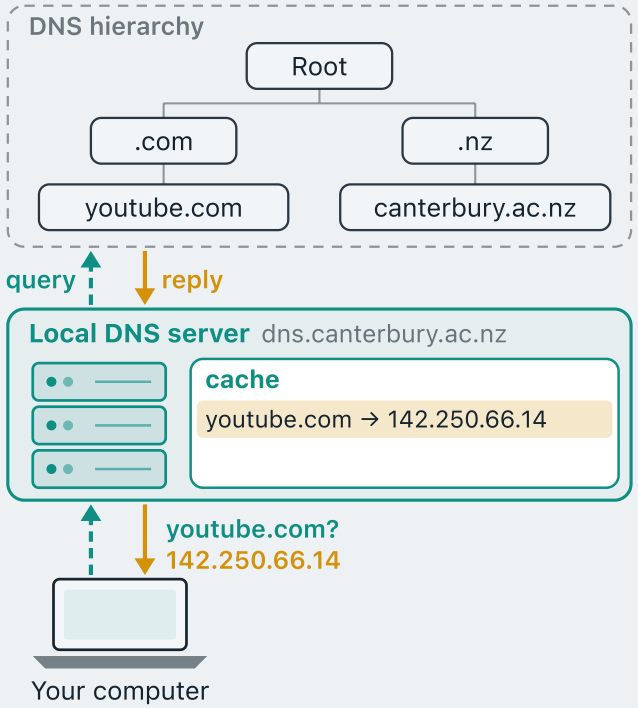
# Local DNS server and DNS caching

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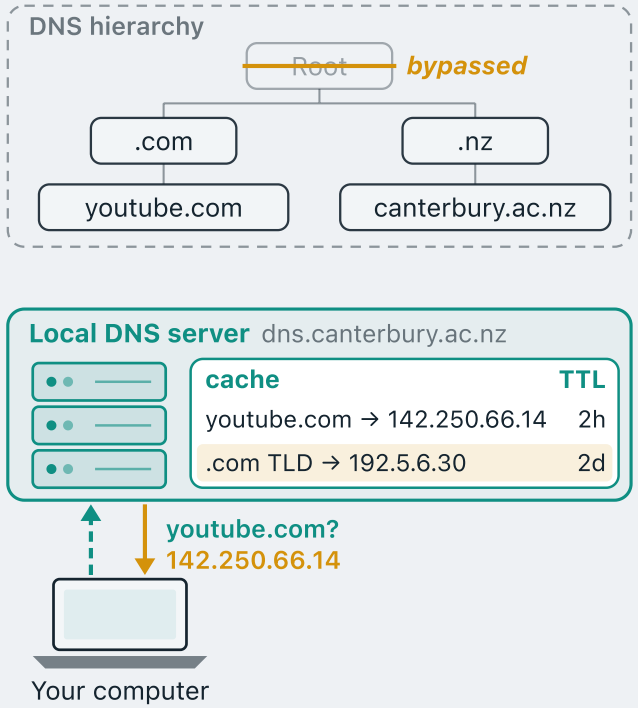
and **caches the hostname-to-IP mapping** for future queries.



DNS ARCHITECTURE

# Local DNS server and DNS caching

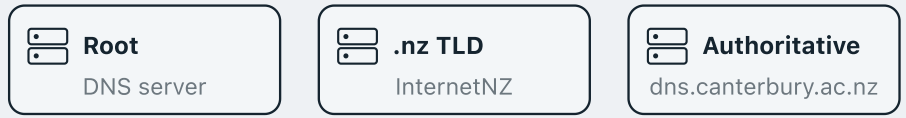
Cache entries have a **TTL (time-to-live)** and expire after some time (typically hours). TLD server addresses are routinely cached in local DNS servers — so **root servers are bypassed for most everyday queries**.

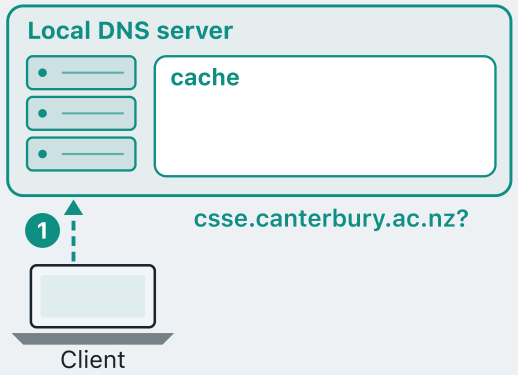


DNS ARCHITECTURE

# Resolving a hostname: the full journey

**Step 1.** A client wants to resolve **csse.canterbury.ac.nz**. It sends a DNS query to its **local DNS server**.

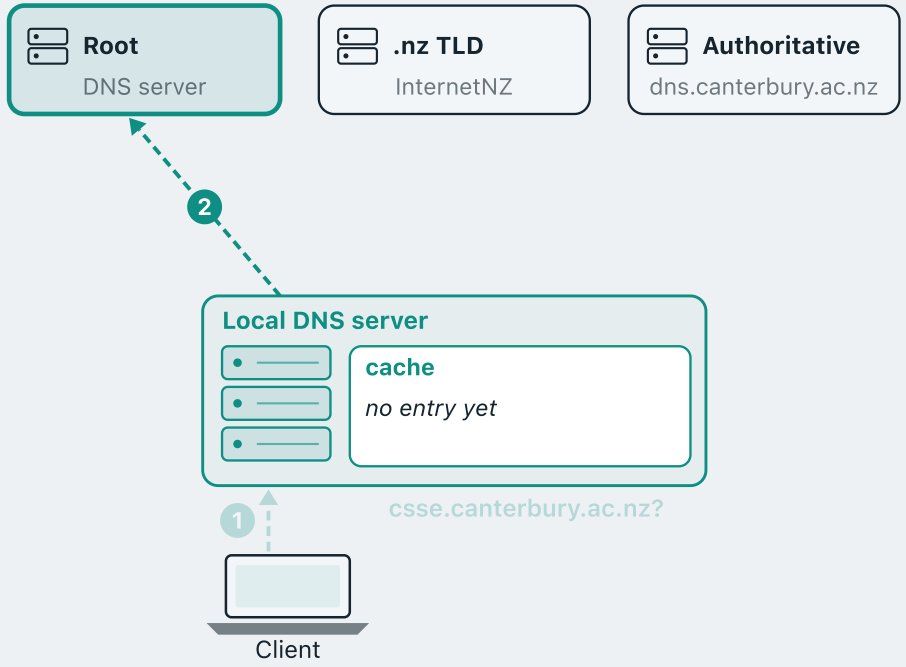


DNS ARCHITECTURE

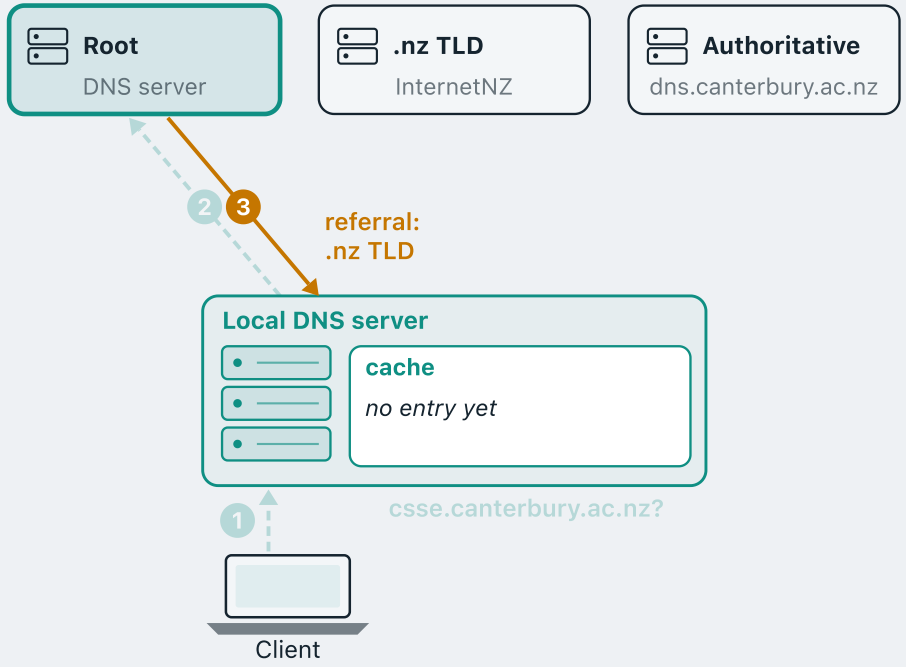
# Resolving a hostname: the full journey

**Step 2.** The local DNS doesn't have the answer cached. It contacts a **root DNS server**.



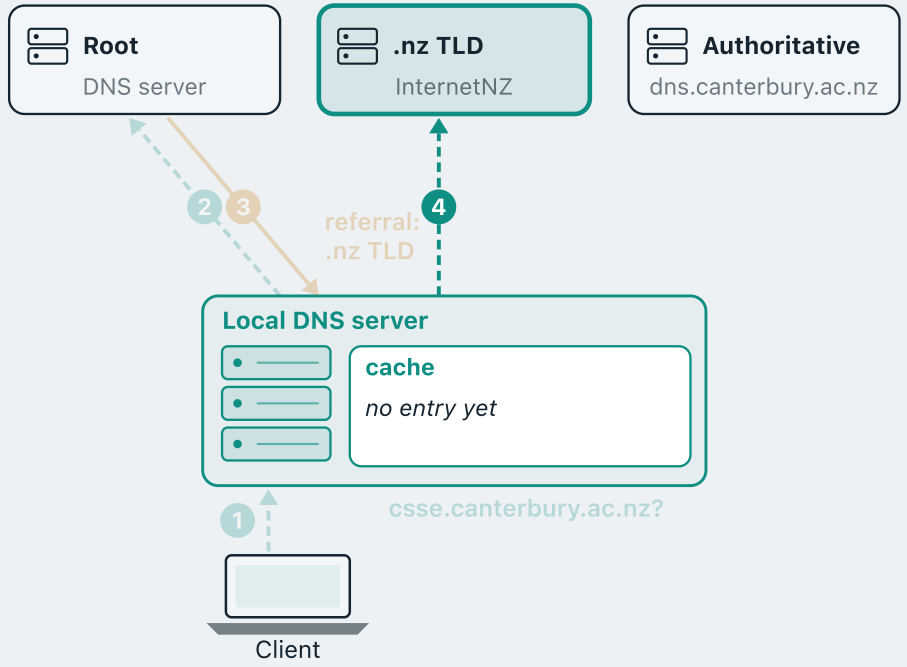
# Resolving a hostname: the full journey

**Step 3.** The root server returns a **referral**: "I don't know — try a **.nz TLD** server" and gives its address.



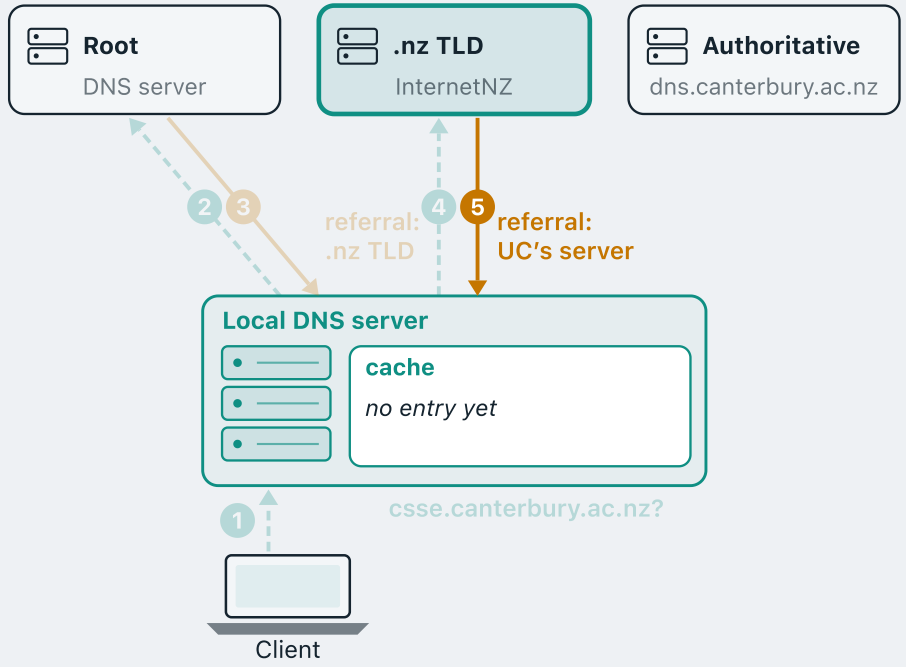
# Resolving a hostname: the full journey

**Step 4.** The local DNS follows the referral and contacts the **.nz TLD server**.



# Resolving a hostname: the full journey

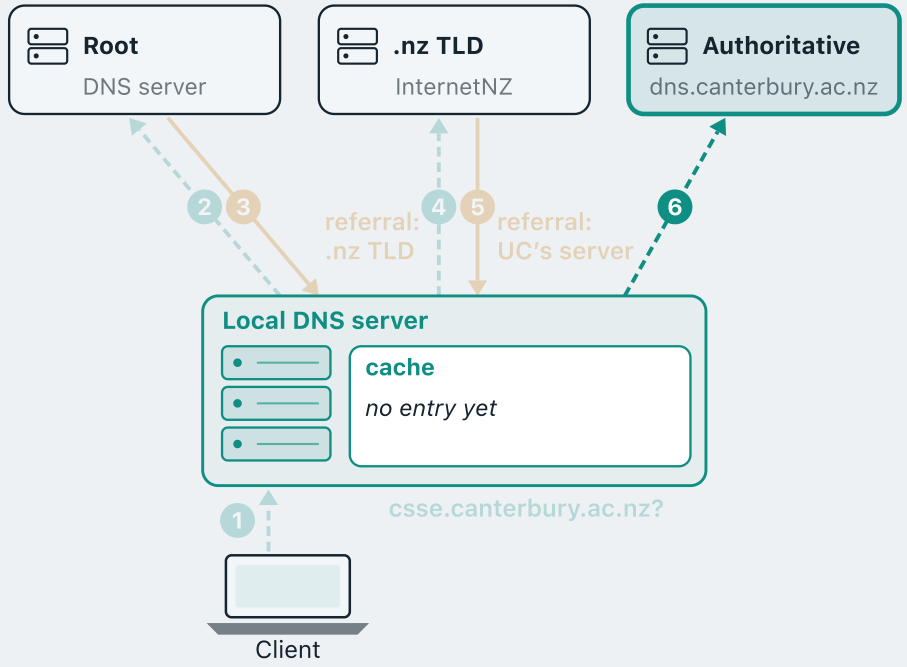
**Step 5.** The .nz TLD returns another **referral**: "Try **dns.canterbury.ac.nz** — the authoritative server for canterbury.ac.nz."



DNS ARCHITECTURE

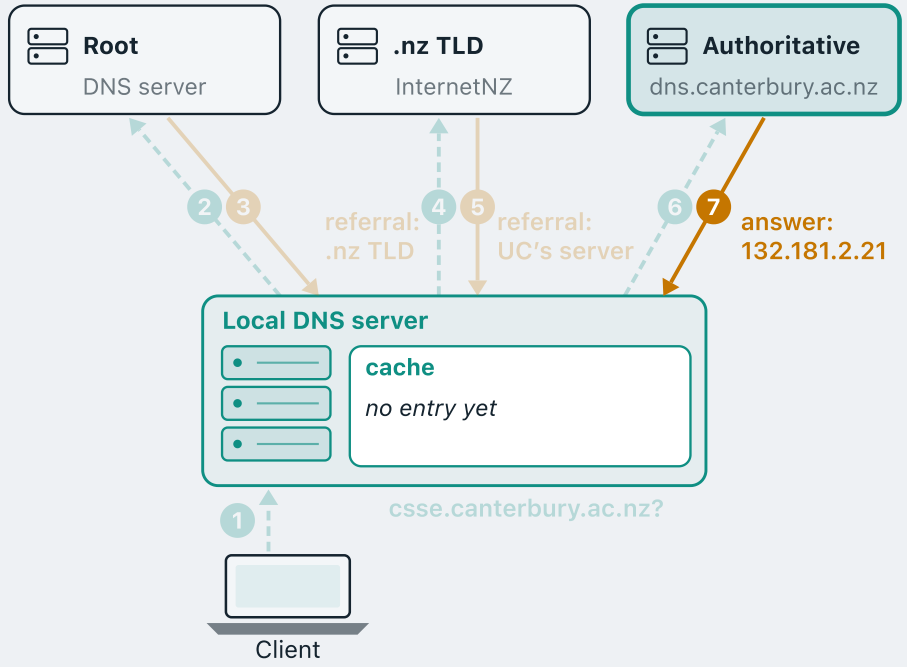
# Resolving a hostname: the full journey

**Step 6.** The local DNS contacts **dns.canterbury.ac.nz** — the authoritative name server for the canterbury.ac.nz zone.



# Resolving a hostname: the full journey

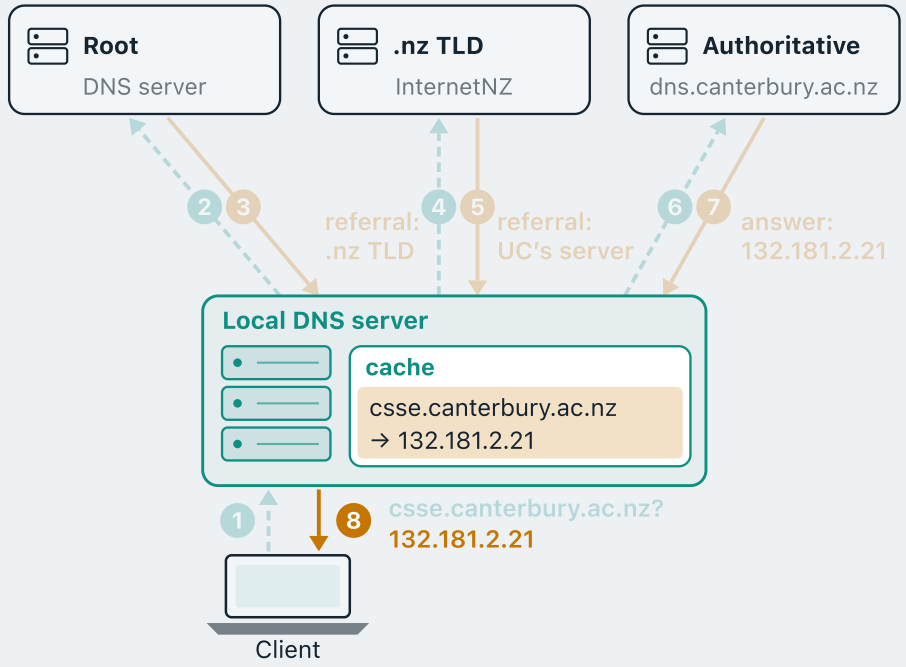
**Step 7.** The authoritative server has the answer! It replies: "csse.canterbury.ac.nz = **132.181.2.21**"



DNS ARCHITECTURE

# Resolving a hostname: the full journey

**Step 8.** The local DNS returns **132.181.2.21** to the client — and **caches** the mapping for future queries.

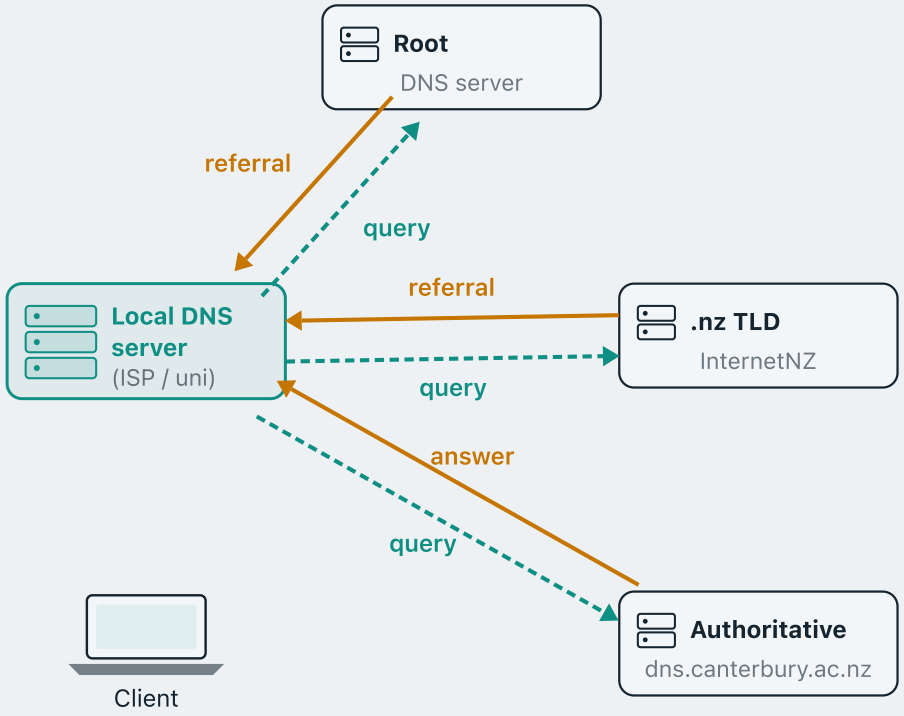


DNS ARCHITECTURE

# Recursive vs iterative queries

### Iterative query

- The contacted server does **not** query other servers for you: it replies only with what it knows itself.
- If it doesn't know the answer, it returns a **referral**: "I don't know — try that server."
- The authoritative server *does* know, so its reply to the same kind of query is the **answer**.

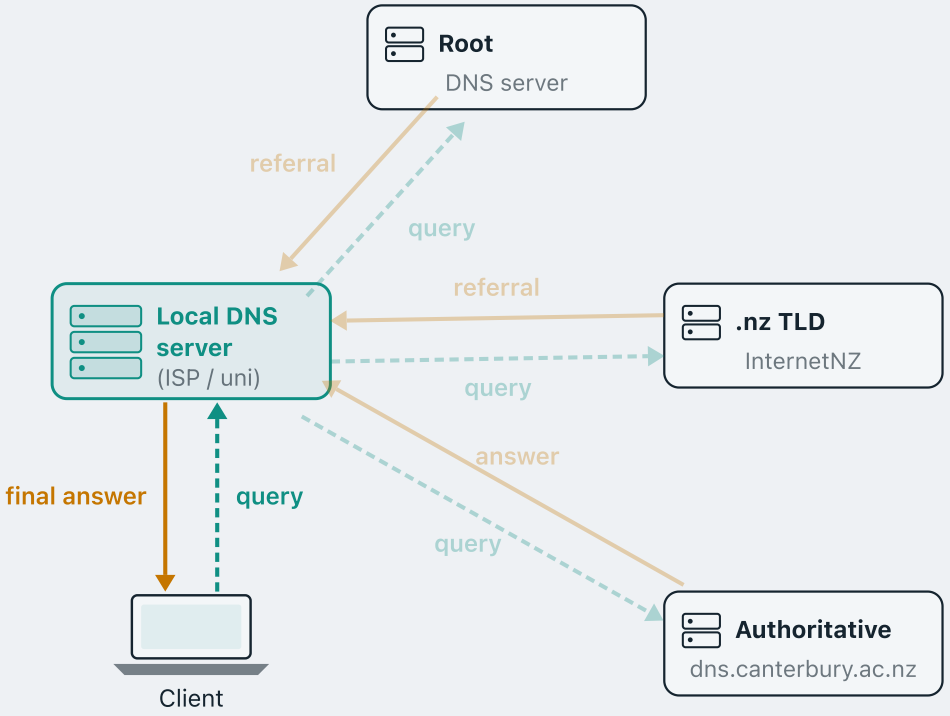


DNS ARCHITECTURE

# Recursive vs iterative queries

**Recursive query**

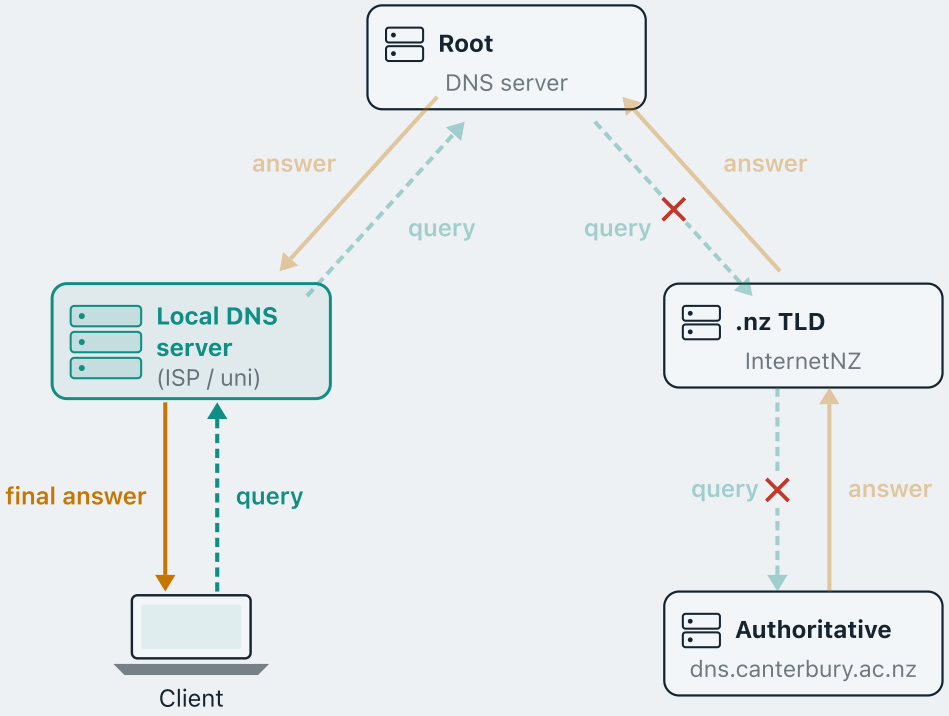
- The contacted server takes **full responsibility**:
- it resolves the name itself, querying the next servers on your behalf,
- and returns the **final answer**. e.g. client → local DNS server.



# Recursive vs iterative queries

**Can we go fully recursive?**

- In principle, yes: root asks the TLD server, which asks the authoritative server,
- and the answer flows back along the chain.
- In practice, root and TLD servers **refuse recursion**: too much load.



DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

**Type A**

**name** = hostname · **value** = IPv4 address

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

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| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

### Type NS

**name** = domain · **value** = hostname of the authoritative DNS server for that domain

*"I don't know — but here's the name of the server that does."*

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

- **name** — e.g. a hostname
- **value** — e.g. an IP address
- **type** — A, NS, CNAME, MX
- **TTL** — time-to-live (s)

### Type CNAME

**name** = alias hostname · **value** = canonical (real) hostname it maps to

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

### Type MX

**name** = domain · **value** = hostname of the mail server for that domain

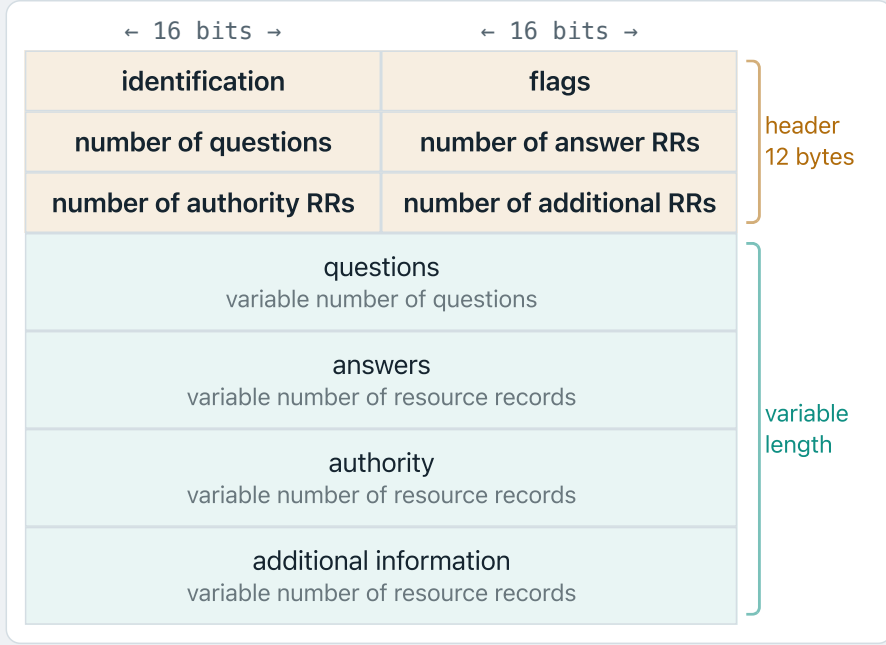
Unlike CNAME, it only redirects mail: a sending mail server (**SMTP**) looks it up to find where to deliver a domain's email.

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS protocol message format

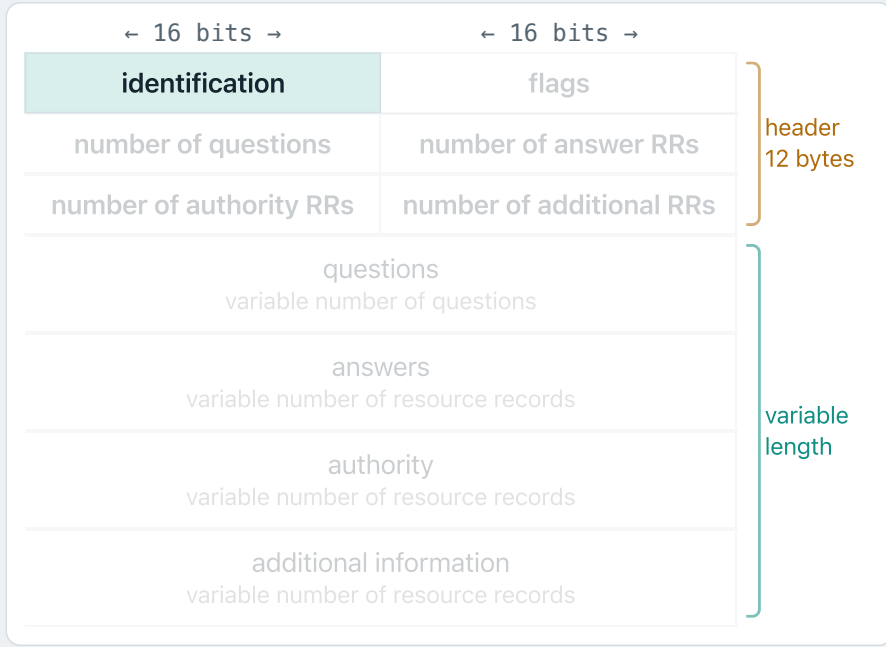
DNS uses the **same message format** for both **queries** and **replies** — a fixed 12-byte header followed by four variable-length sections.



# DNS protocol message format

**Identification**

16-bit number assigned by the client. The reply copies this same value, so the client can match each reply to its original query.

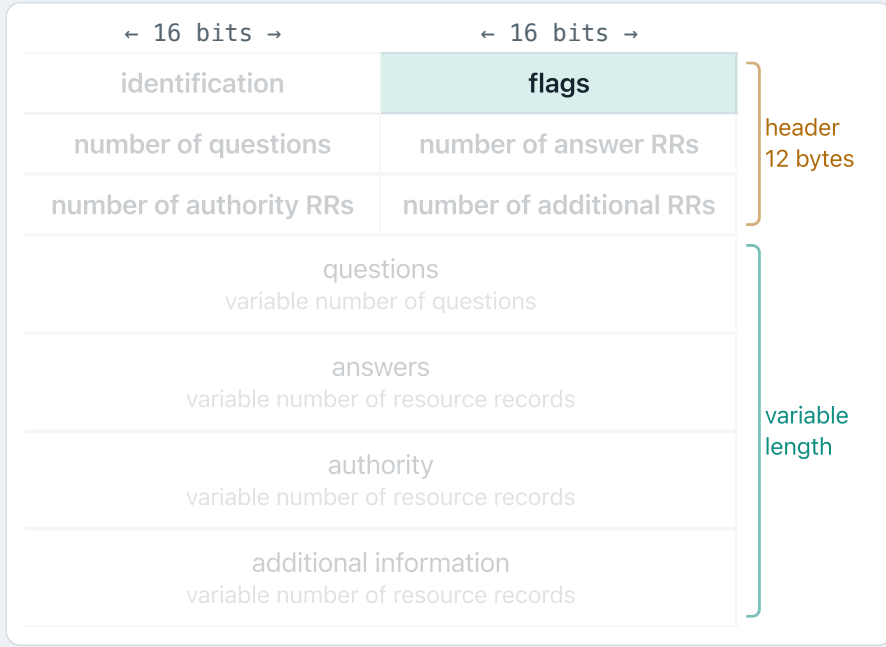


DNS ARCHITECTURE

# DNS protocol message format

**Flags**

- Query (0) or reply (1)
- Recursion desired (RD) — client requests full recursive resolution
- Recursion available (RA) — server indicates it supports recursion

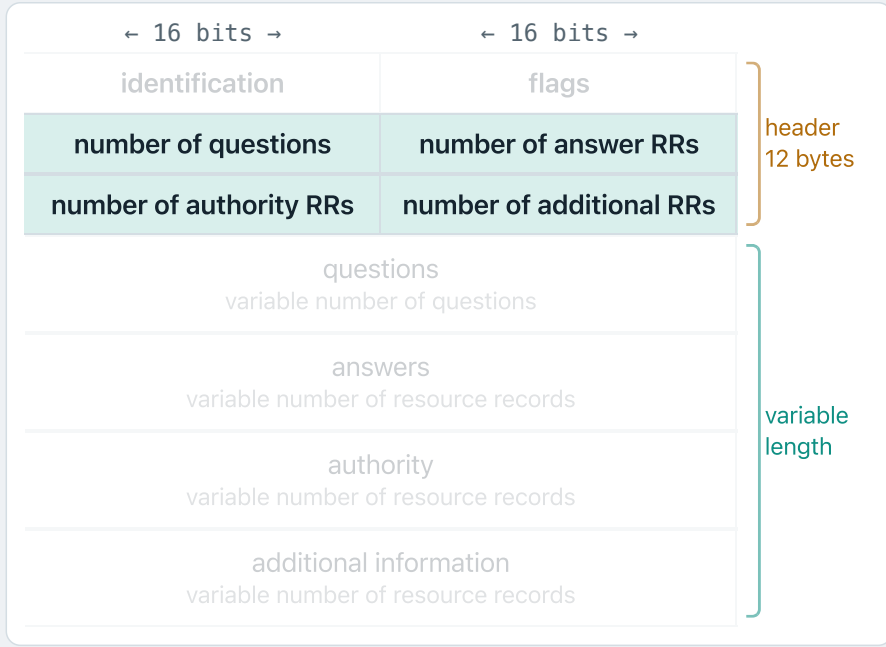


DNS ARCHITECTURE

# DNS protocol message format

### Count fields

Four 16-bit counters — one per section below — telling how many records appear in each variable-length section.

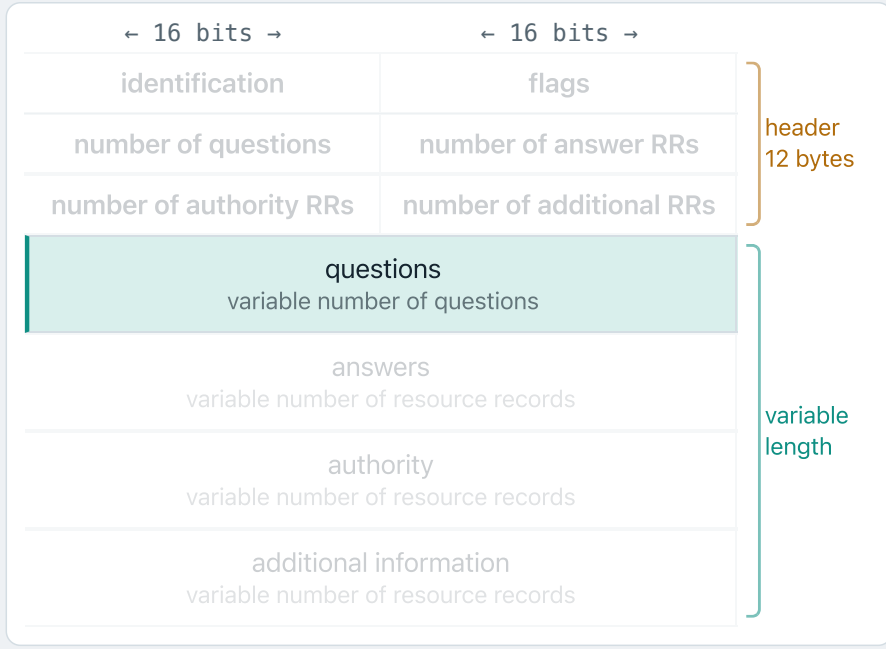


# DNS protocol message format

### Questions

The name and record type being queried. e.g. this Type A question asks for the IP address of the domain facebook.com.

```
(facebook.com, A)
```

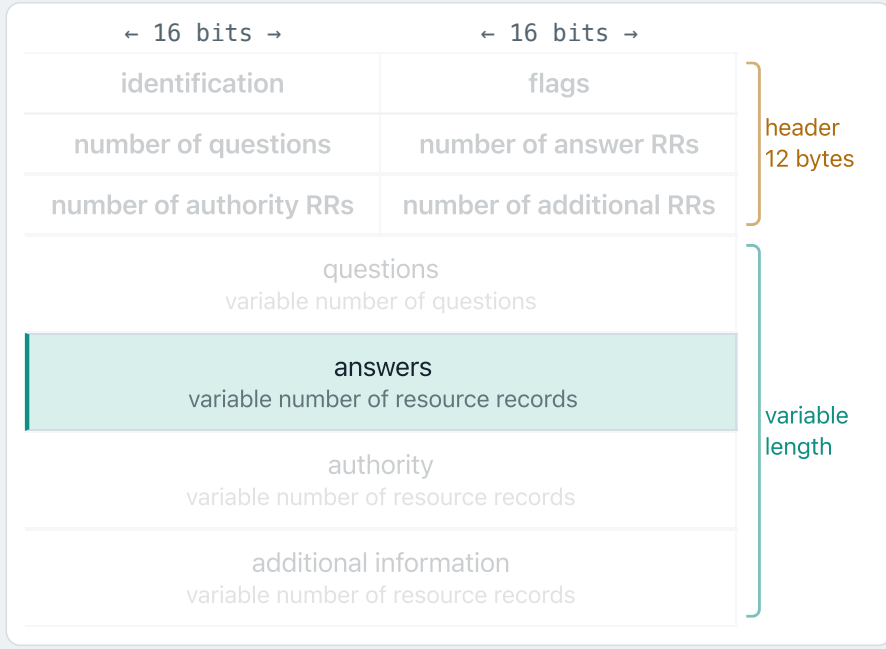


# DNS protocol message format

**Answers**

Resource Records (RRs) that directly answer the query.

```
(facebook.com, 157.240.22.35, A, TTL=300)
```

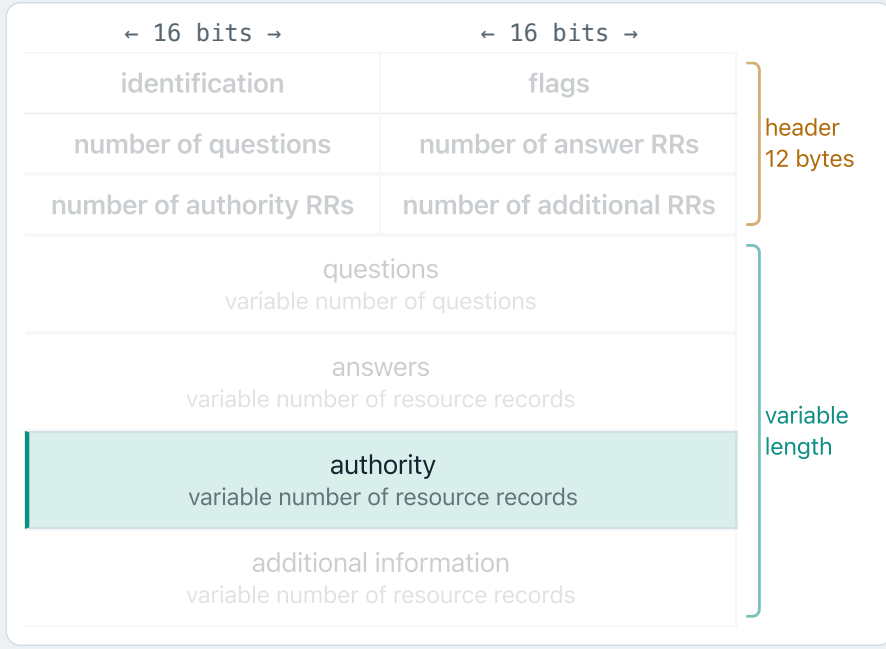


# DNS protocol message format

**Authority**

NS records pointing to authoritative servers — returned in referral responses.

```
(facebook.com, a.ns.facebook.com, NS)
```

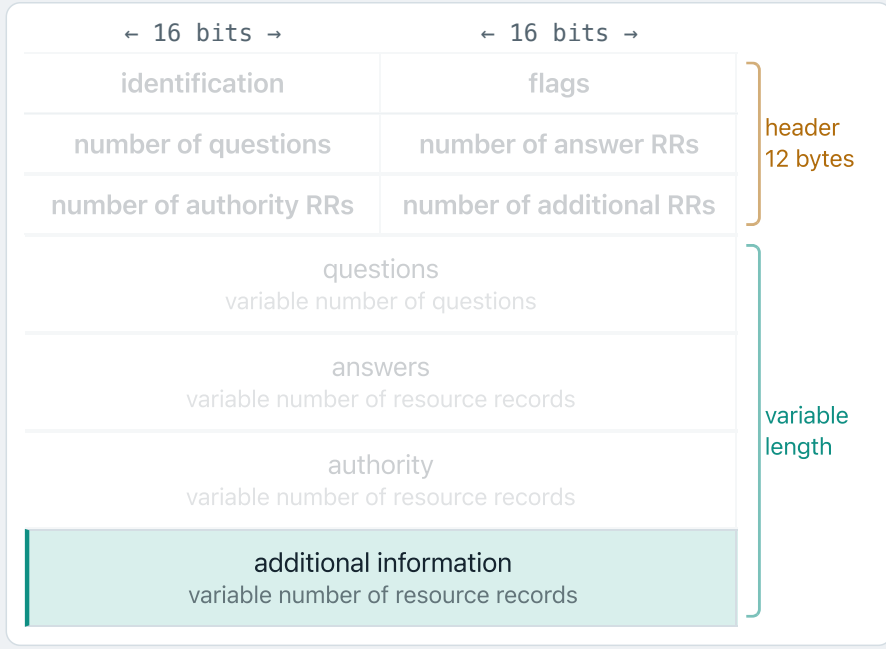


# DNS protocol message format

**Additional**

'Glue' Resource Records (RRs) accompanying the authority section — so you can reach the name server without an extra lookup.

```
(a.ns.facebook.com, 129.134.30.12, A)
```



# Query & response: an example

**QUERY** — client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

### The question

The query asks one question: what is the IP address (type **A**) of `www.canterbury.ac.nz`?

**RESPONSE** — local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**The answer**

The response answers with **two records**.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**First: a CNAME**

`www.canterbury.ac.nz` is an **alias**; its canonical name is `web.canterbury.ac.nz`.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**Then: an A record**

The canonical name's IP address, 132.181.106.9 — the answer we wanted.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

### Authority

An **NS** record naming an authoritative server for `canterbury.ac.nz`: the final level, which holds the answer's record.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** — client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**Additional**

The **IP address** of that name server, so it can be reached directly.

**RESPONSE** — local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

DNS

# DNS — key takeaways

- **Main purpose** — DNS maps human-readable hostnames to IP addresses.
- **Hierarchy** — DNS is distributed: no single server holds all records.
  Responsibility is delegated across Root, TLD, and Authoritative servers.
- **Traversal pattern** — the client queries the local DNS resolver, which then follows referrals Root → TLD → Authoritative.
  Results are cached locally by TTL to avoid repeated lookups.

# References


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William Stallings, *Data and Computer Communications*, Pearson, 10th edition, 2013.

Chapter 24 Electronic Mail, DNS, and HTTP (24.1–24.3)

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5th edition, 2010.

Chapter 7 The Application Layer (7.1–7.3)

# Acknowledgements

These lecture notes are developed based on slides and notes from:

Barry Wu's lecture notes for COSC264, University of Canterbury

Dr Mengmeng Ge's lecture notes for COSC264, University of Canterbury, 2024

Assoc Prof Dan Kim's lecture notes for COSC264, University of Canterbury, 2017

Prof Aleksandar Kuzmanovic's lecture notes for CS340, Northwestern University, 2005